写出下列程序的运行结果

```
int main()
{  const char *c[]={"John learn C++ language",
                    "Be well!", "You", "Not very"};
   const char **p[]={c+3, c+2, c+1, c};
   const char ***pp=p;
   cout << (**++pp);
   cout << (*--*++pp+4);
   cout << (*pp[-2]+3);
   cout << (pp[-1][-1]+2);
   cout << endl;
   return 0;
}
```

注：直接在本文件上作答，画出程序执行过程的内存变化即可

★ 首先画出三句定义语句结束后内存中各变量的所占空间及初值

★ 每个执行语句的每一步执行完成后的内存中各变量的所占空间及值

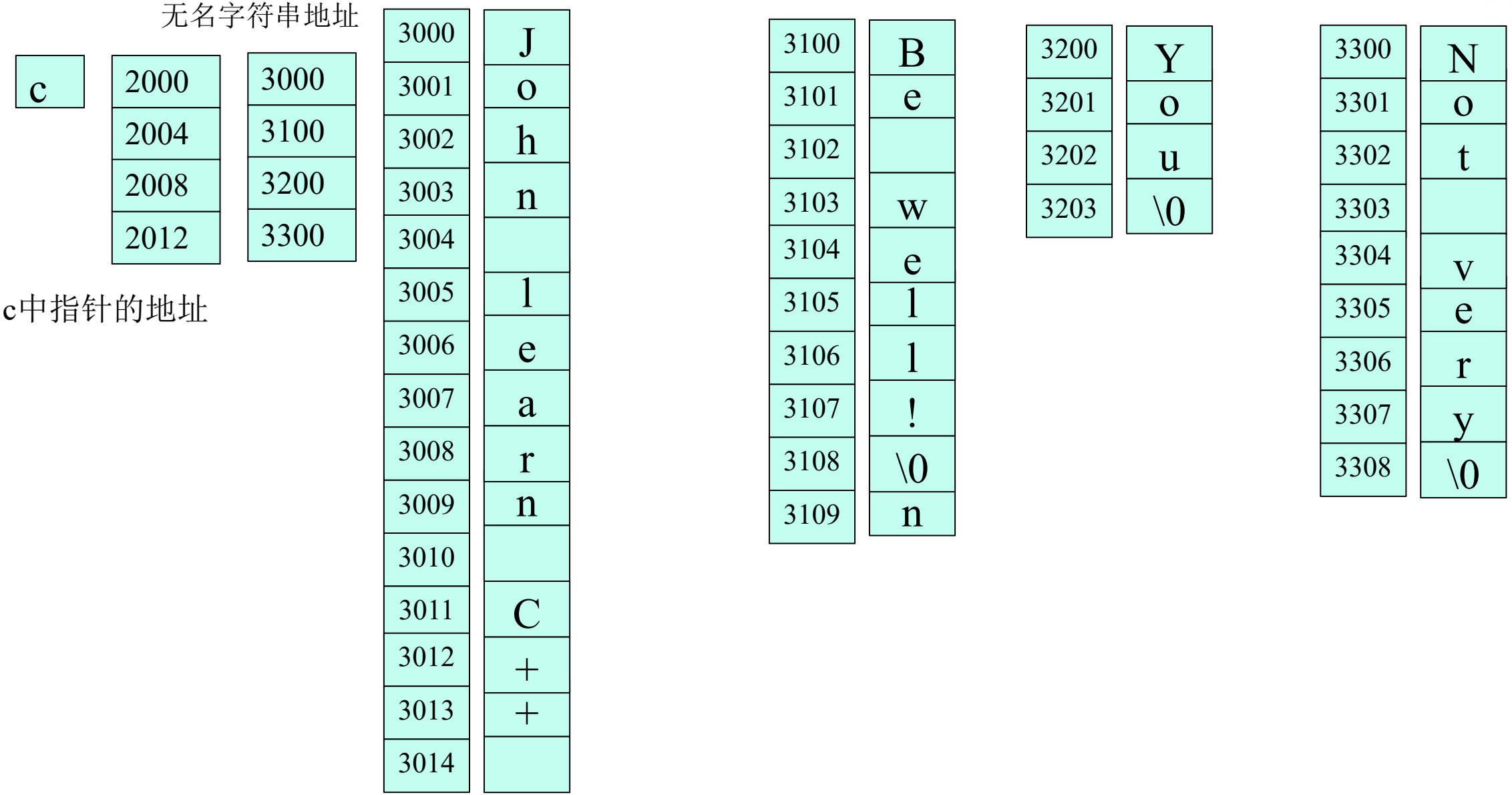
★ 每步变化一个页面(例：**++pp，分三步计算，需要三页)

★ 不允许手写在纸上，再拍照贴图

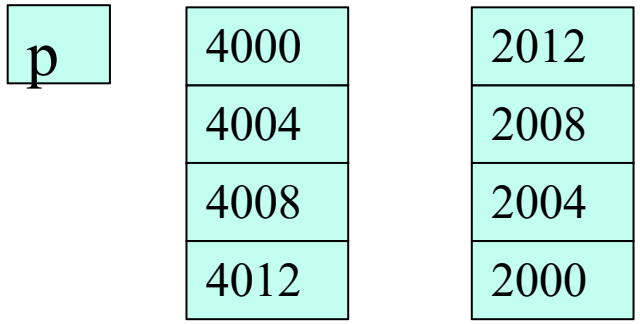
★ 允许在各种软件工具上完成，再截图贴图

★ 转换为pdf后提交

```
const char *c[]={"John learn C++ language",
                 "Be well!", "You", "Not very"};
```



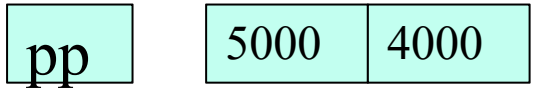
```
const char **p[]={c+3, c+2, c+1, c};
```



p中的地址　　对应的c指针地址

```
const char ***pp=p;
```

其实是const char** (*pp)=p



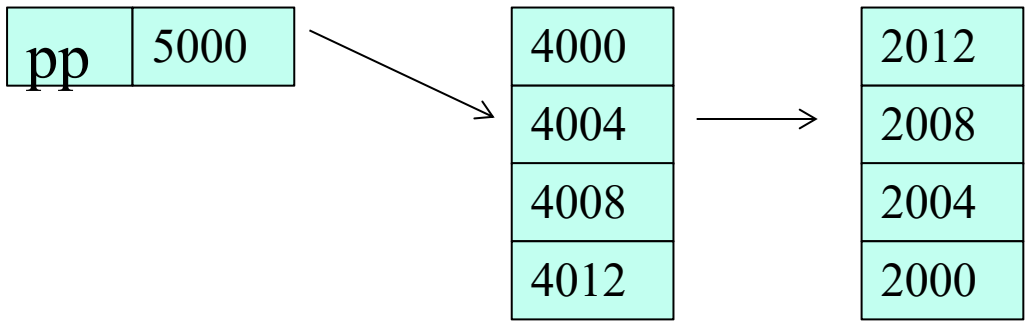
pp地址　　pp指向p首元素地址

**++pp

先进行++操作，pp指向p[1]

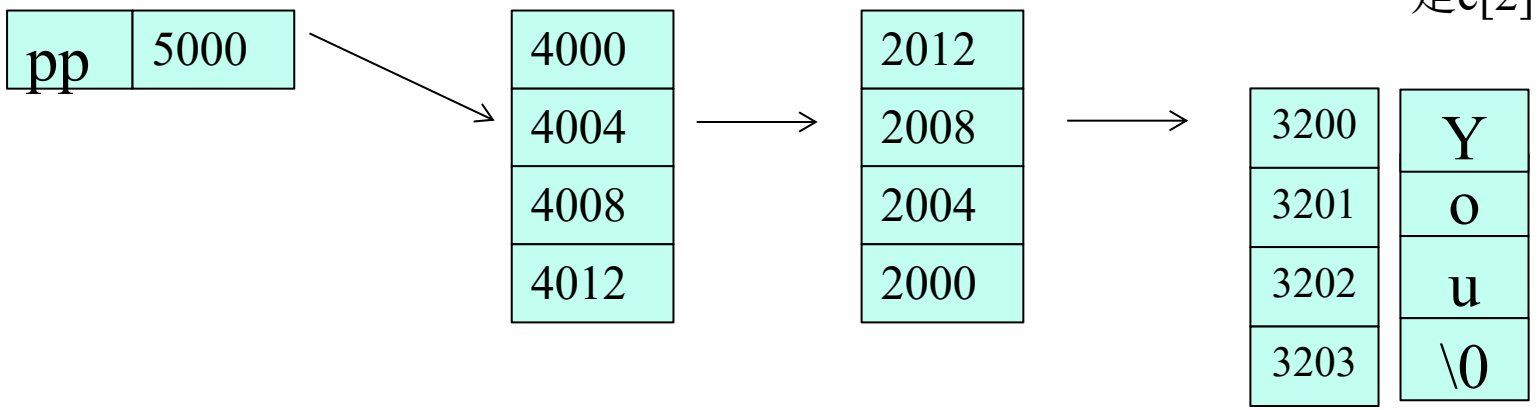
pp	5000	4004

**++pp



第二步，取p[1]的值，也就是c[2]的地址

**++pp



第三步，取c[2]的值，也就是c[2]的地址

因此输出“You”

(*--*++pp+4)

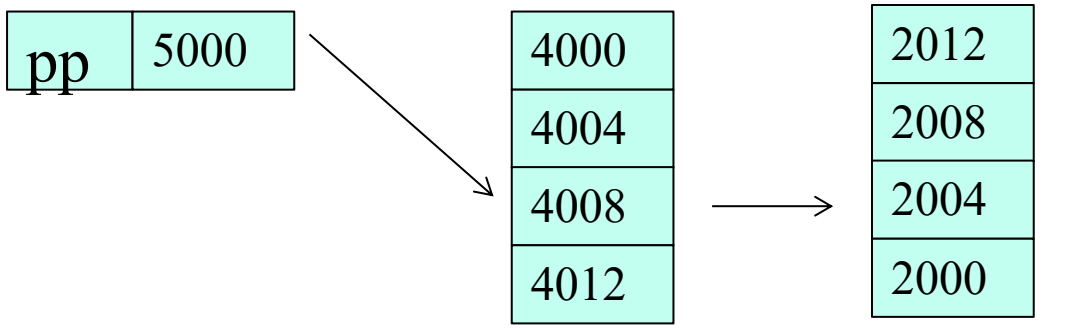
第一步，++pp,指向p[2]

pp	5000	4008

(*--*++pp+4)

const char **p[3]

第二步，取*，得到p[2]地址

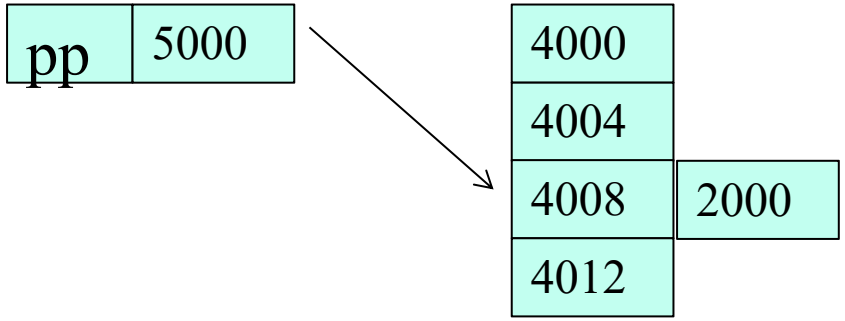


const char *c

(*--*++pp+4)

const char **p[3]

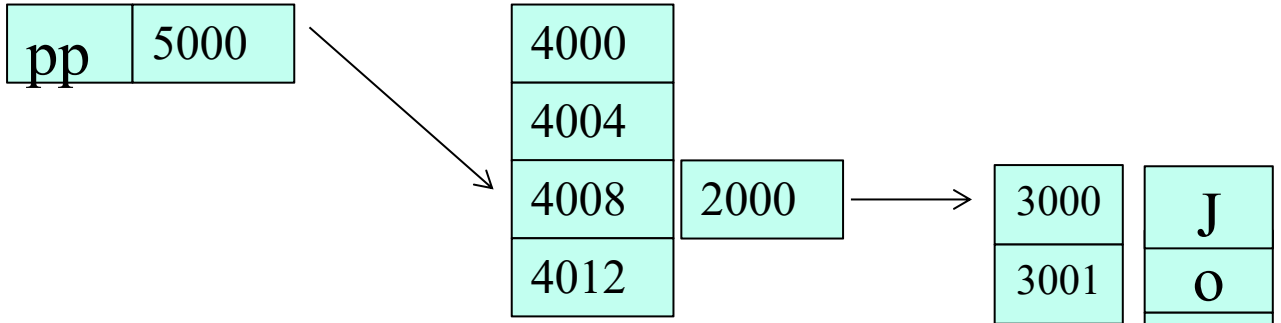
第三步，--，p[2]指向2000



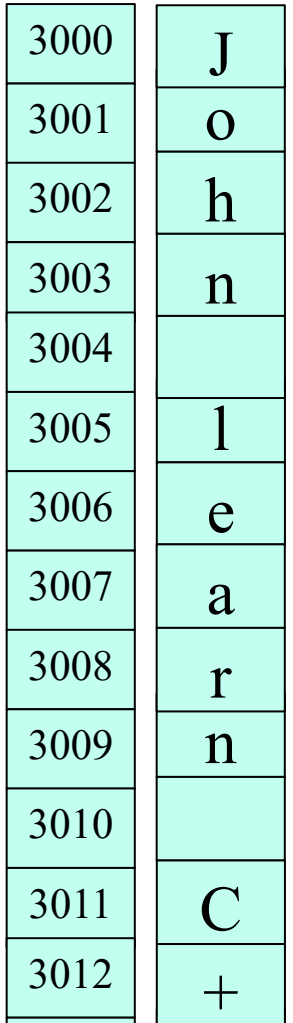
(*--*++pp+4)

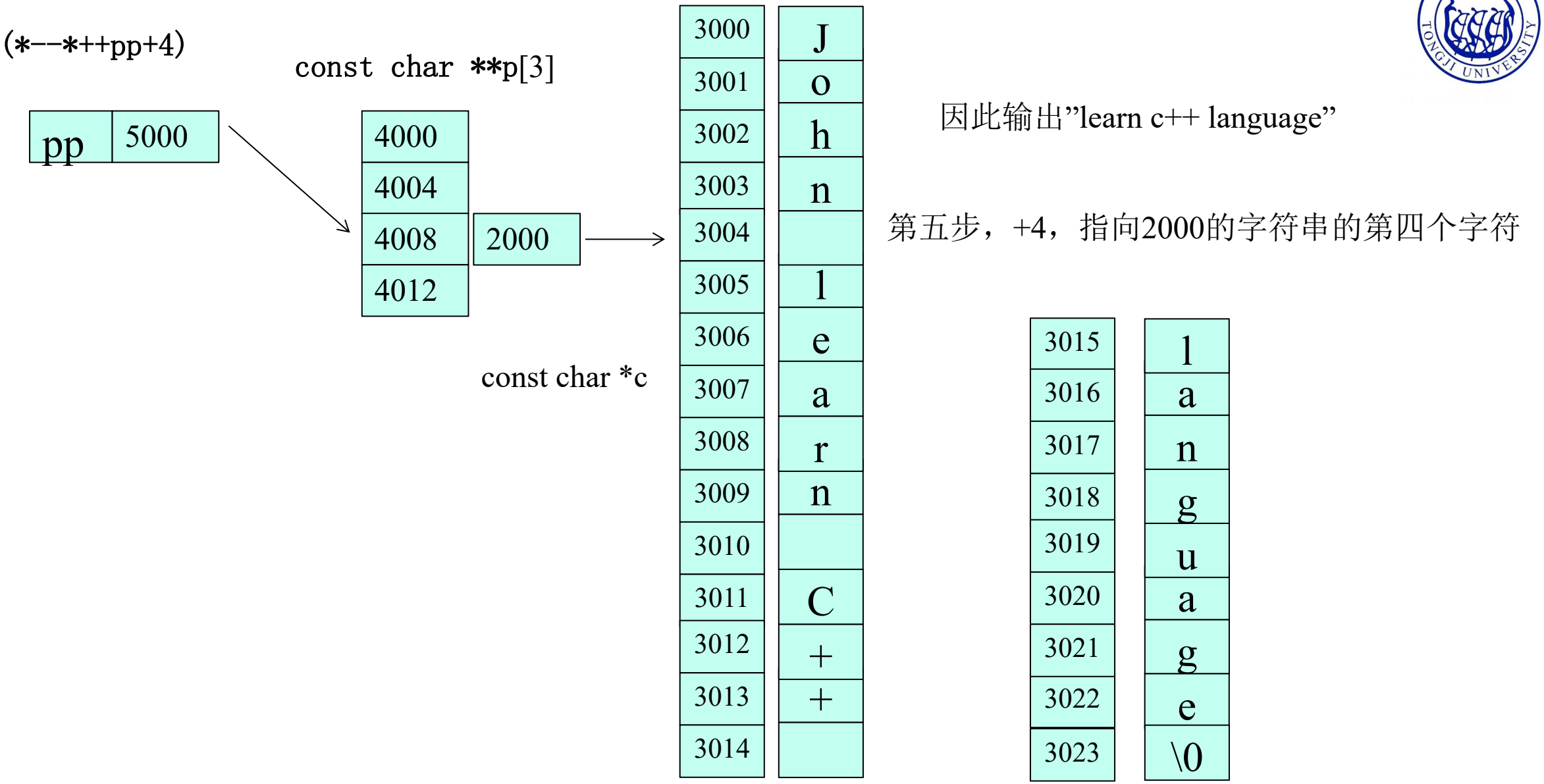
const char **p[3]

第四步，取*c[0]，此时位置是c[0]



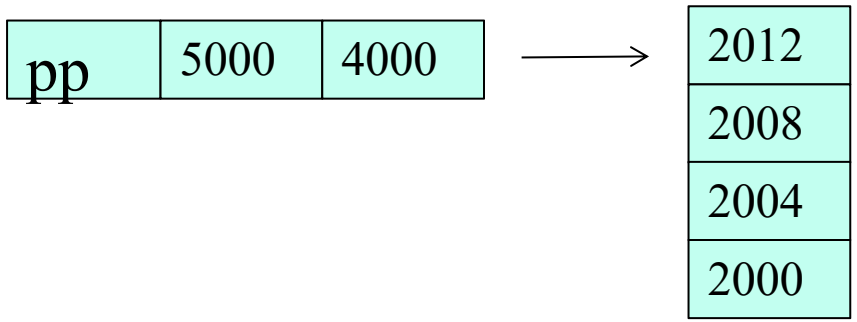
const char *c




(*--*++pp+4)
const char **p[3]
pp
5000
4000
4004
4008
2000
4012
const char *c
3000
3001
3002
3003
3004
3005
3006
3007
3008
3009
3010
3011
3012
3013
3014
J
o
h
n
l
e
a
r
n
C
+
+
因此输出”learn c++ language”
第五步，+4，指向2000的字符串的第四个字符
3015
3016
3017
3018
3019
3020
3021
3022
3023
l
a
n
g
u
a
g
e
\0

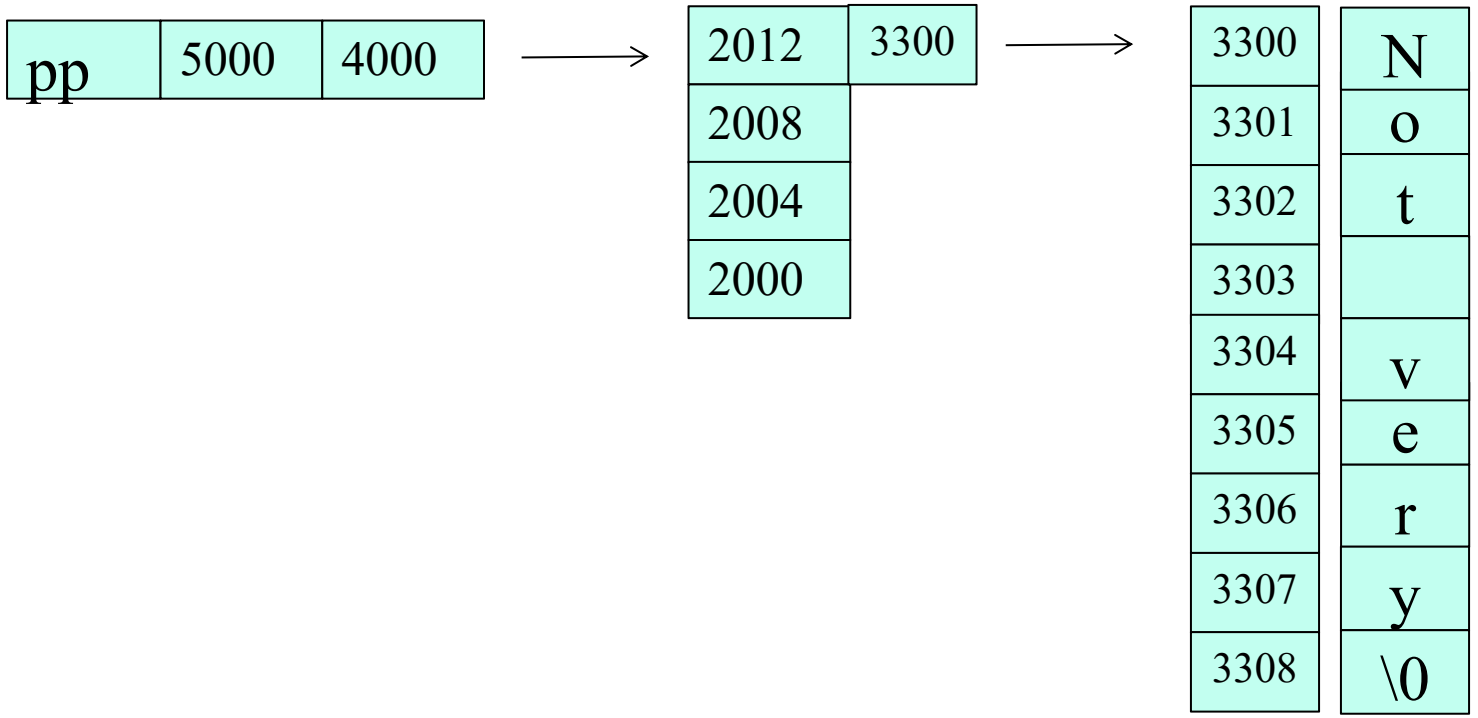
cout << (*pp[-2]+3);

第一步，pp原先存有p[2]的地址，pp[-2]则即是*（pp-2）



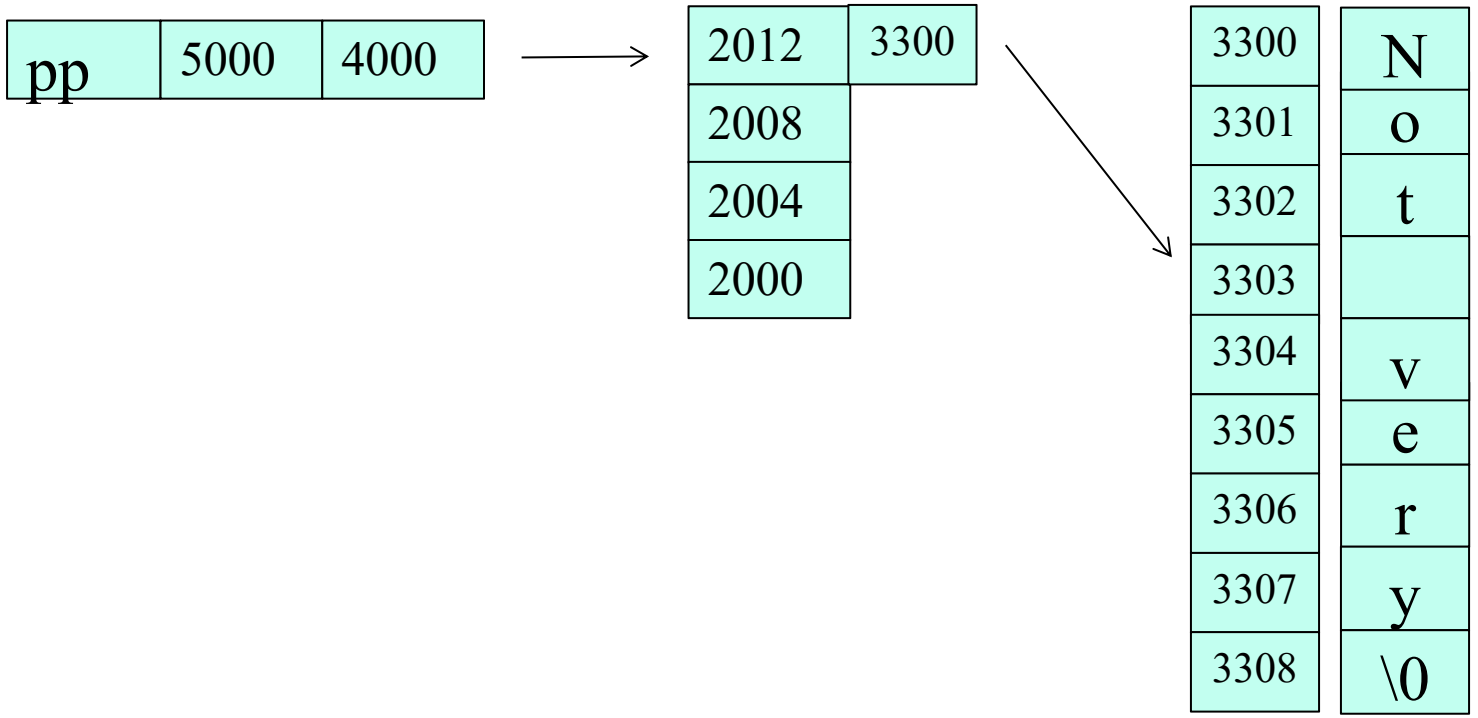
cout << (*pp[-2]+3);

第二步，取*,也就是取2012指针里存的3300位置的字符串



cout << (*pp[-2]+3);

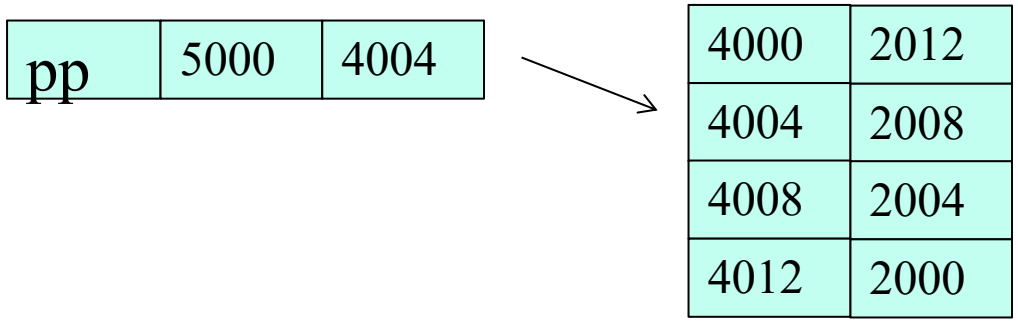
第三步，+3,此时的位置是3303



输出“ very”

cout << (pp[-1][-1]+2);

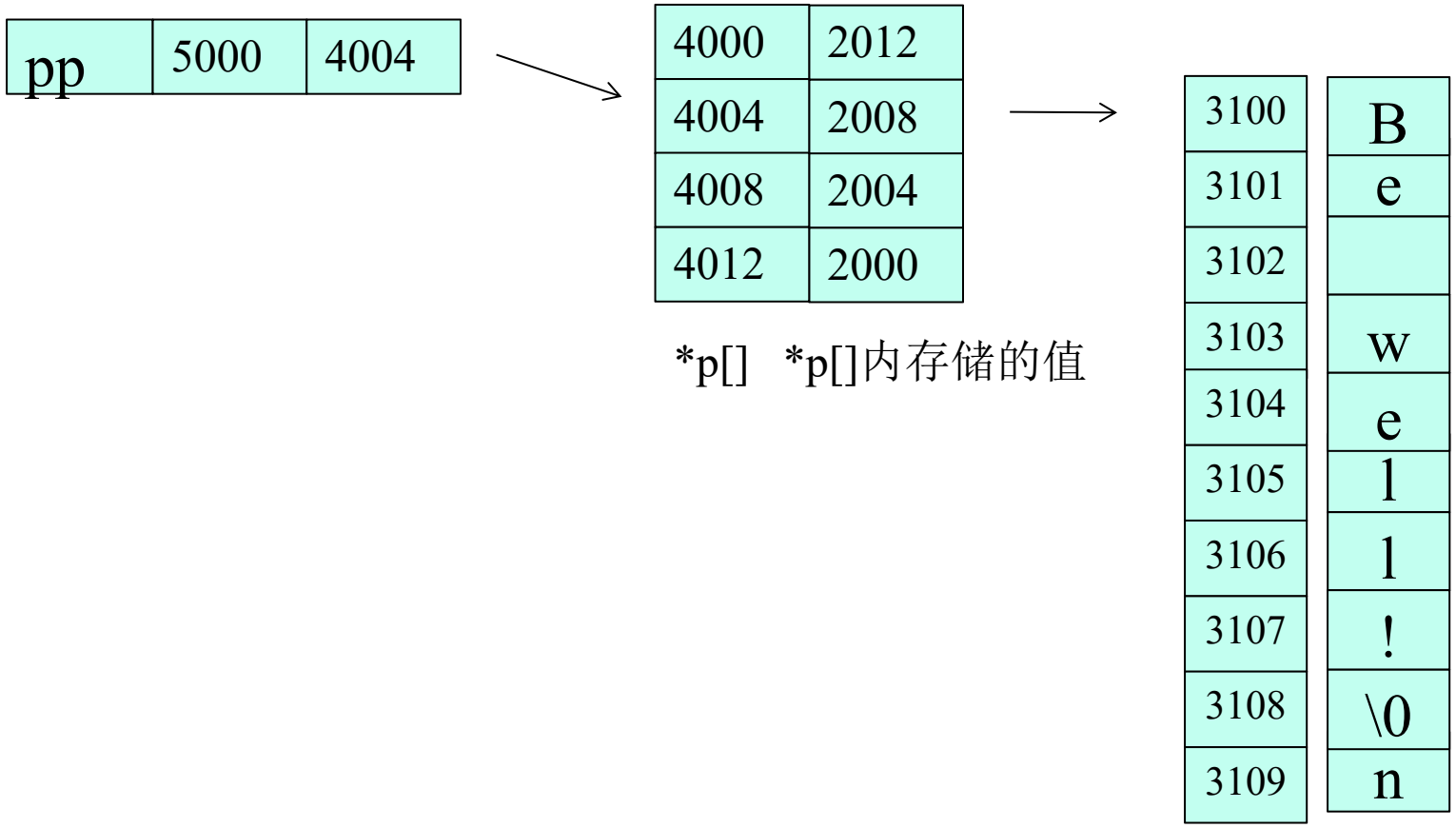
第一步，pp[-1],也就是*(pp-1),取的是4004的地址



*p[]　*p[]内存储的值

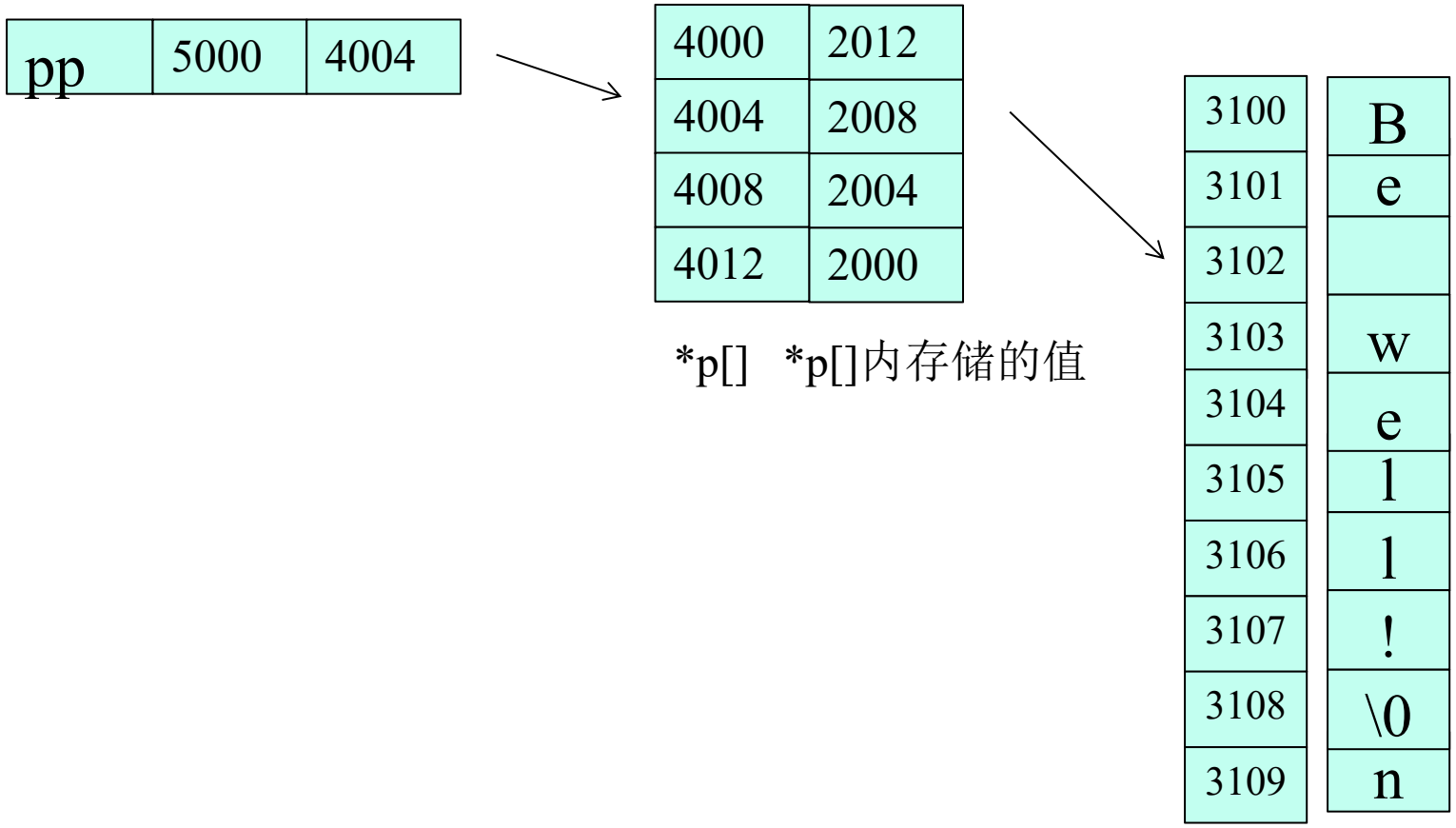
cout << (pp[-1][-1]+2);

第二步，pp[-1][-1],也就是*（*(pp-1)-1）,取到c[1]字符串



cout << (pp[-1][-1]+2);

第三步，+2,也就是从3102开始



输出“ well!”

最后输出：“You learn c++ language very well!”